CSS for Styling

The good, the bad and the… ugly!


```
<p>
<font face="Arial">Shashdot.</font>
News for <b>nerds!!</b> You will <i>never</i>, <u>EVER</u>
be
<font size="+4" color="red">BORED</font> here!
</p>
```

HTML

- Why is this bad?

Cascading Style Sheets (CSS)

- Describes the appearance, layout, and presentation of information on a web page
  - HTML describes **the content** of the page
- Describes *how* information is to be displayed, not *what* is being displayed
- Can be embedded in HTML document or placed into separate .css file

Inline styles: the style attribute


```
<p style="font-family: sans-serif; color: red;">
This is a paragraph</p>
```

HTML

This is a paragraph

output

- Higher precedence than embedded or linked styles
- Used for one-time overrides and styling a particular element
- Bad *style* and should be avoided when possible (why?)

Basic CSS rule syntax


```
selector {
property: value;
property: value;
...
property: value;
}
```
CSS

```
p {
font-family: sans-serif;
color: red;
}
```
CSS

- A CSS file consists of one or more **rules**
- Each rule starts with a **selector**
- A selector specifies an HTML element(s) and then applies style **properties** to them
  - a selector of * selects all elements

Embedding style sheets: <style>


```HTML
<head>
<style type="text/css">
p { font-family: sans-serif; color: red; }
h2 { background-color: yellow; }
</style>
</head>
```

- CSS code can be embedded within the head of an HTML page
- B*ad style* and should be avoided when possible (why?)

Attaching a CSS file <link>


```
<head>
...
<link href="filename" type="text/css" rel="stylesheet" />
...
</head>
```

HTML

```
<link href="style.css" type="text/css" rel="stylesheet" />
<link href="http://www.google.com/uds/css/gsearch.css"
rel="stylesheet" type="text/css" />
```

HTML

- A page can link to multiple style sheet files
  - In case of a conflict (two sheets define a style for the same HTML element), the latter sheet's properties will be used

CSS properties for colors


```css
p {
color: red;
background-color: yellow;
}
```

CSS

This paragraph uses the style above

output

property	description
color	color of the element's text
background-color	color that will appear behind the element

Specifying colors


```
p { color: red; }
h2 { color: rgb(128, 0, 196); }
h4 { color: #FF8800; }
```

CSS

This paragraph uses the first style above

This h2 uses the second style above.

This h4 uses the third style above.

output

- color names: aqua, black, blue, fuchsia, gray, green, lime, maroon, navy, olive, purple, red, silver, teal, white (white), yellow
- RGB codes: red, green, and blue values from 0 (none) to 255 (full)
- hex codes: RGB values in base-16 from 00 (0, none) to FF (255, full)

Grouping styles


```
p, h1, h2 {
color: green;
}
h2 {
background-color: yellow;
}
```

CSS

This paragraph uses the above style.

This h2 uses the above styles.

output

- A style can select multiple elements separated by commas
- The individual elements can also have their own styles

CSS comments /*...*/


```
/* This is a comment.
It can span many lines in the CSS file. */
p {
color: red; background-color: aqua;
}
```
CSS

- CSS (like HTML) is usually not commented as rigorously as programming languages such as Java
- The // single-line comment style is NOT supported in CSS
- The <!-- ... --> HTML comment style is also NOT supported in CSS

CSS properties for fonts

property	description
font-family	which font will be used
font-size	how large the letters will be drawn
font-style	used to enable/disable italic style
font-weight	used to enable/disable bold style

Complete list of font properties (https://developer.mozilla.org/en-US/docs/Web/CSS/font)

font-family


```
p {
font-family: Georgia;
}
h2 {
font-family: "Courier New";
}
```

CSS

This paragraph uses the first style above.

This h2 uses the second style above.

output

- Enclose multi-word font names in quotes

More about font-family


```
p {
font-family: Garamond, "Times New Roman", serif;
}
```
CSS

This paragraph uses the above style.

output

- We can specify multiple fonts from highest to lowest priority
- Generic font names:
  - serif, sans-serif, cursive, FANTASY, monospace
- If the first font is not found on the user's computer, the next is tried
- Placing a generic font name at the end of your font-family value, ensures that every computer will use a valid font

font-size


```
p {
        font-size: 24pt;
}
```
CSS

This paragraph uses the style above.

output

- units: pixels (`px`) vs. point (`pt`) vs. m-size (`em`)

`16px`, `16pt`, `1.16em`

- vague font sizes: `xx-small`, `x-small`, `small`, `medium`, `large`, `x-large`, `xx-large`, `smaller`, `larger`
- percentage font sizes, e.g.: `90%`, `120%`

font-size


```
p {
        font-size: 24pt;
}
```
CSS

This paragraph uses the style above.

output

- `pt` specifies number of point, where a point is 1/72 of an inch onscreen
- `px` specifies a number of pixels on the screen
- `em` specifies number of m-widths, where 1 em is equal to the font's current size

font-weight, font-style


```
p {
font-weight: bold;
font-style: italic;
}
```

CSS

This paragraph uses the style above.

output

- Either of the above can be set to normal to turn them off (e.g. headings)

CSS properties for text

property	description
text-align	alignment of text within its element
text-decoration	decorations such as underlining
line-height, word-spacing, letter-spacing	gaps between the various portions of the text
text-indent	indents the first letter of each paragraph

Complete list of text properties
(https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_text)

text-align


```
blockquote { text-align: justify; }
h2 { text-align: center; }
```

CSS

The Gollum's Quote

We wants it, we needs it. Must have the precious. They stole it from us. Sneaky little hobbitses. Wicked, tricksy, false!

output

- text-align can be `left`, `right`, `center`, or `justify`



text-decoration


```
p {
text-decoration: underline;
}
```
CSS

This paragraph uses the style above.

output

- can also be `overline`, ~~`line-through`~~, `blink`, or `none`
- effects can be combined:

```
text-decoration: overline underline;
```

The list-style-type property


```
ol { list-style-type: lower-roman; }
```
CSS

- Possible values:

  i. `none` : No marker

  ii. `disc` (default), `circle`, `square`

  iii. `Decimal`: 1, 2, 3, etc.

  iv. `decimal-leading-zero`: 01, 02, 03, etc.

  v. `lower-roman`: i, ii, iii, iv, v, etc.

  vi. `upper-roman`: I, II, III, IV, V, etc.

  vii. `lower-alpha`: a, b, c, d, e, etc.

  viii. `upper-alpha`: A, B, C, D, E, etc.

  x. `lower-greek`: alpha, beta, gamma, etc.

  others: hebrew, armenian, georgian, cjk-ideographic, hiragana…

Body styles


```css
body {
font-size: 16px;
}
```

CSS

- Applies a style to the entire body of your page
- Saves you from manually applying a style to each element

Cascading Style Sheets

- Properties of an element *cascade* together in this order:
  - browser's default styles
  - external style sheet files (in a <link> tag)
  - internal style sheets (inside a <style> tag in the page's header)
  - inline style (the style attribute of the HTML element)

Inheriting styles


```
body { font-family: sans-serif; background-color: yellow;
}
p { color: red; background-color: aqua; }
a { text-decoration: underline; }
h2 { font-weight: bold; text-align: center; }
```

CSS

This is a heading

A styled paragraph. Previous slides are available on the website.

- A bulleted list

output

- when multiple styles apply to an element, they are inherited
- a more tightly matching rule can override a more general inherited rule

Styles that conflict


```
p, h1, h2 { color: blue; font-style: italic; }
h2 { color: red; background-color: yellow; }
```

CSS

This paragraph uses the first style above.

This heading uses both styles above.

output

- when two styles set conflicting values for the same property, the latter style takes precedence

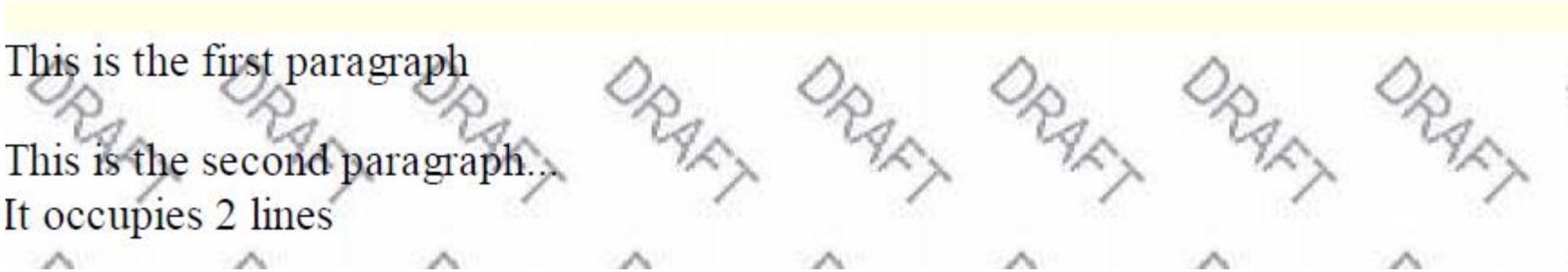
CSS properties for backgrounds

property	description
background-color	color to fill background
background-image	image to place in background
background-position	placement of bg image within element
background-repeat	whether/how bg image should be repeated
background-attachment	whether bg image scrolls with page
background	shorthand to set all background properties

background-image


```
body {
background-image: url("images/draft.jpg");
}
```

CSS



- background image/color fills the element's content area

background-repeat


```
body {
background-image: url("images/draft.jpg");
background-repeat: repeat-x;
}
```

CSS



- can be repeat (default), repeat-x, repeat-y, or no-repeat

background-position


```
body {
background-image: url("images/draft.jpg");
background-repeat: no-repeat;
background-position: 370px 20px;
}
```
CSS

This is the first paragraph

This is the second paragraph...
It occupies 2 lines



- value consists of two tokens, each of which can be top, left, right, bottom, center, a percentage, or a length value in px, pt, etc.
- value can be negative to shift left/up by a given amount

Exercise

- Define styles for tag (e.g., div), class, and id.
- Show the privilege